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# Re-Implementing CNN Price-Trend Signals: A Reproduction of Jiang, Kelly, and Xiu (2023)

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## Abstract

This paper reproduces the image-based price-trend prediction framework of Jiang et al. [2023], which replaces predefined technical indicators with OHLC-volume charts and convolutional neural networks (CNNs). Using CRSP daily U.S. equity data, we reconstruct the image pipeline, train horizon-specific CNN ensembles, and evaluate the resulting signals through out-of-sample backtests. We compare the CNN signals with momentum, moving-average, and reversal benchmarks, and assess residual performance through factor-spanning regressions. The reproduction is strongest at short horizons, where image-based signals deliver the best economic performance. Relative to the magnitudes reported by Jiang et al. [2023], medium- and long-horizon strategies weaken substantially after costs, and factor stripping yields significantly negative alphas. Saliency maps indicate that the CNN mainly attends to recent price movement and volume contrast. We release all code and artefacts needed to audit and extend the reproduction.

## 1 Introduction

Technical analysis has long suggested that the visual “shape” of a price chart conveys information beyond what is captured by scalar momentum or moving-average filters. Jiang et al. [2023] formalise this intuition by feeding fixed-resolution open–high–low–close (OHLC) images with a volume sub-panel into a convolutional neural network and showing that the resulting forecasts dominate canonical price-trend signals out of sample. Their work joins a growing empirical asset-pricing literature that uses flexible machine-learning predictors to map raw historical features into expected returns [Gu et al., 2020, Kelly et al., 2019], and it ties directly to early efforts at formalising chart patterns via kernel smoothing [Lo et al., 2000].

This paper re-implements the core pipeline from end to end and evaluates it under the same in-sample and out-of-sample calendar design as Jiang et al. [2023]: training and validation draw from 1993–2000 with a 70/30 random sample split, and headline out-of-sample results are reported for 2001–2019. Our contributions are:

1. An independent OHLC-plus-volume image renderer and horizon-matched CNN stacks built from the dimensions, drawing rules, and layer geometry documented in Jiang et al. [2023]—including the simple moving-average overlay, volume sub-panel, and vertical orientation convention—implemented from the paper’s written specification and reported tensor dimensions.
2. A CRSP-style daily panel spanning 1992–2024 for data construction, with out-of-sample CNN evaluation for 2001–2019 as above; proportional transaction costs; matched-horizon price momentum, simple moving-average cross, and short-term reversal baselines; and a five-factor-plus-momentum spanning regression [Fama and French, 1993, 2015, Carhart, 1997].

3. Input-gradient saliency analysis [Simonyan et al., 2014] of the  $I = 60$  ensemble, which highlights the right tail of each image and the volume bars as the dominant feature regions.
4. Public release of code, trained weights, and post-aggregation CSV tables in `final_submit_version/`, allowing every numerical claim to be regenerated.

The remainder of the paper is organised as follows. Section 2 positions our reproduction relative to the chart-pattern, deep-learning, and interpretability literatures. Section 3 details data sources and image construction. Section 4 describes the CNN architecture and training recipe. Section 5 reports empirical results (decile prediction accuracy, baseline comparisons, cumulative returns, classification metrics, decile portfolio statistics, and Fama–French spanning regressions). Section 6 visualises model attention. Section 7 discusses where our results align with and diverge from Jiang et al. [2023].

## 2 Related work

**Chart-pattern literature.** Lo et al. [2000] provided one of the first systematic treatments of technical chart patterns by combining kernel regression with automated pattern recognition. Jegadeesh and Titman [1993] documented price momentum as a robust cross-sectional return predictor, providing the canonical benchmark against which any technical-trend signal must be compared.

**Deep learning for return prediction.** Recent work demonstrates that deep architectures uncover return signals that are difficult to capture with closed-form characteristics. Gu et al. [2020] benchmark a broad set of supervised learners on U.S. equities; Kelly et al. [2019] use latent-factor models with characteristic loadings. Within technical analysis, Sezer and Ozbayoglu [2018] convert indicators into images for trading. Jiang et al. [2023] unify these threads by treating the OHLC chart itself as the model input.

**Interpretability via input gradients.** Simonyan et al. [2014] introduced gradient-based saliency to visualise which input pixels most influence a CNN’s class score; we apply the same technique to identify which segments of each OHLC image drive the model’s “up” probability. CNN architectures used here trace back to LeCun et al. [1998].

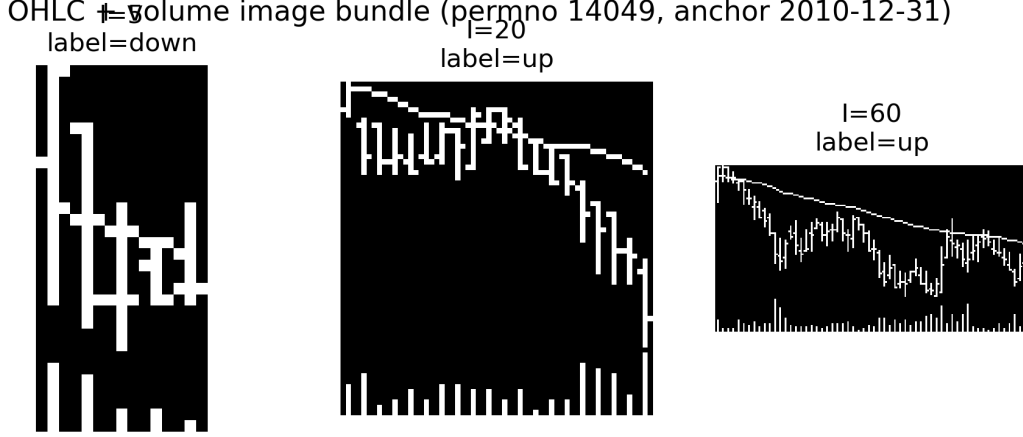
## 3 Data and image construction

**Universe.** We use a CRSP-style daily equity panel covering 1992-01-01 to 2024-12-31. The cleaning step retains the following columns per `final_submit_version/docs/variable_description.txt`: `permno`, `dlycaldt`, `dlycap`, `dlyret`, `dlyretx`, `dlyvol`, `dlyopen`, `dlyhigh`, `dlylow`, `dlyclose`. We drop rows with any of those columns missing, apply a \$1 minimum-close and strictly positive volume filter, and compute log returns and adjusted OHLC prices via cumulative log-return compounding so that the first valid observation per `permno` acts as the price anchor. No survivorship adjustments are applied beyond what is implicit in the CRSP-style file.

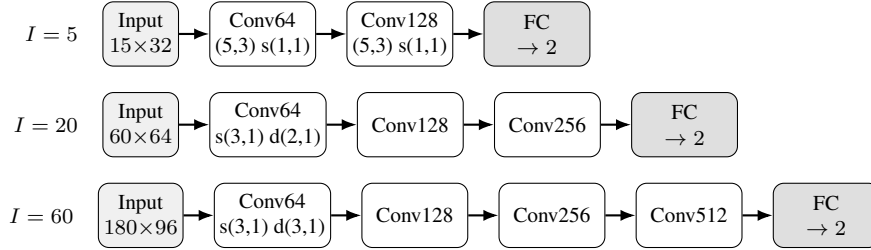
**Labels and holding period.** For each horizon  $F \in \{5, 20, 60\}$  trading days we form  $R_{t,F}^{\text{fut}} = \exp(\sum_{s=t+1}^{t+F} \log(1 + r_s)) - 1$  on the adjusted price path, and a binary classification label  $y_{t,F} = \mathbb{1}\{R_{t,F}^{\text{fut}} > 0\}$ . Rebalance timestamps are fixed at period-end calendar boundaries (Friday for  $F = 5$ , month-end for  $F = 20$ , quarter-end for  $F = 60$ ), matching the canonical SPY trade calendar.

**Image construction.** For input horizon  $I \in \{5, 20, 60\}$  matched to  $F = I$  we render a grayscale image of width  $W = 3I$  and height  $H$  split into an OHLC panel of height  $H_{\text{ohlc}}$  stacked above a one-pixel-gap volume panel of height  $H_{\text{vol}}$ :

| Horizon  | $W$ | $H_{\text{ohlc}}$ | $H_{\text{vol}}$ |
|----------|-----|-------------------|------------------|
| $I = 5$  | 15  | 25                | 6                |
| $I = 20$ | 60  | 51                | 12               |
| $I = 60$ | 180 | 76                | 19               |



**Figure 1:** OHLC-plus-volume images for a single security at horizons  $I \in \{5, 20, 60\}$ . The same calendar anchor (2010-12-31) is used so the visual density at different resolutions is directly comparable.



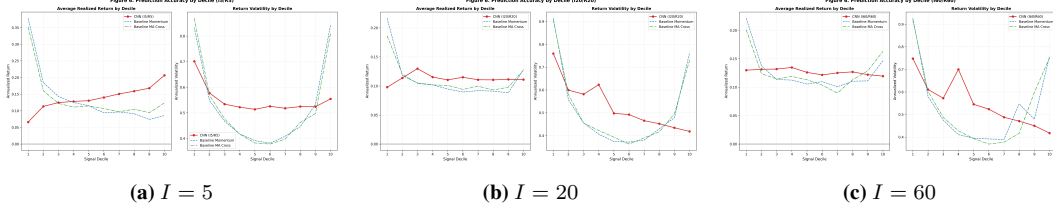
**Figure 2:** Horizon-specific CNN backbones. Each rectangle represents one Conv→BatchNorm→LeakyReLU→MaxPool block. All kernels are  $5 \times 3$ ; default stride and dilation are (1, 1) except where explicitly noted. The fully-connected head has dropout 0.5 and outputs two logits.

For each day  $\tau$  in the window we draw (a) a one-pixel vertical bar from the low to the high, (b) a one-pixel open tick at column  $3\tau$  and a close tick at column  $3\tau + 2$ , (c) a one-pixel SMA poly-line connecting the rolling  $I$ -day means of close prices (computed using  $2I - 1$  days of history so that the first SMA point already reflects  $I$  observations), all scaled to a common  $[P_{\min}, P_{\max}]$  range that includes the SMA values. Volume bars are normalised to the window’s own maximum. The finished OHLC panel is finally flipped top-to-bottom so that “up” moves appear visually upward, then concatenated with the volume panel with a one-pixel gap. Figure 1 shows the three resolutions for a single permno anchored at the same calendar date.

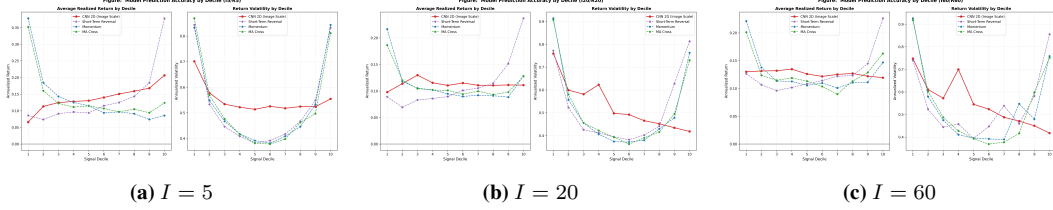
**Train/validation/test split.** We follow Jiang et al. [2023] in treating the early portion of the panel as in-sample. Training is performed on 1993–2000 with a 70/30 random sample-level split for training/validation. Out-of-sample backtests start in 2001 and run through to the end of the cached panel. For the headline tables and figures we report results from 2001-01 to 2019-12 to align with where all baseline ETFs (used for calendar construction) have continuous coverage.

## 4 CNN architecture and training

**Architectures.** The three horizon-specific backbones mirror the published configuration. All blocks use  $5 \times 3$  kernels, a LeakyReLU(0.01) activation, batch normalisation, and (2, 1) MaxPool with `ceil_mode=True` (the  $I = 5$  head uses the standard floor mode).  $I = 20$  and  $I = 60$  insert dilated and strided first convolutions so the receptive field can span longer windows without an explosion in parameter count. Figure 2 summarises the three stacks; layer-level details are in Table 4 (Appendix B).



**Figure 3:** Annualised decile returns by horizon for the CNN, matched-horizon momentum, and moving-average cross signals. Reproduction of Figure 6 of Jiang et al. [2023]. Numbers are read from `outputs/pipeline_runs/tables/fig6_I{5,20,60}_{cnn,mom,ma}.csv`.



**Figure 4:** Simplified Figure 7 reproduction: CNN deciles against momentum, moving-average cross, and short-term reversal baselines. At  $I = 5$  the reversal baseline’s D10 reaches an annualised return of 37.78%, against which the CNN’s monotone ranking is more useful as a signal than as a stand-alone strategy.

**Optimisation.** We minimise binary cross-entropy on the per-horizon label  $y_{t,F}$  using stochastic gradient descent (Adam variant disabled), with learning rate  $10^{-5}$ , batch size 128, and a maximum of 50 epochs with early-stopping on validation loss. Convolutional and linear weights are initialised with Xavier-uniform and biases at zero. We train an ensemble of five seeds  $\{42, 123, 456, 789, 101112\}$  and aggregate by averaging the softmax-probability outputs at inference time. The resulting checkpoints are stored under `final_submit_version/outputs/models/` with sizes  $\approx 0.6$  MB ( $I = 5$ ),  $\approx 2.8$  MB ( $I = 20$ ), and  $\approx 11.8$  MB ( $I = 60$ ) per seed, consistent with the parameter counts implied by Figure 2.

## 5 Empirical results

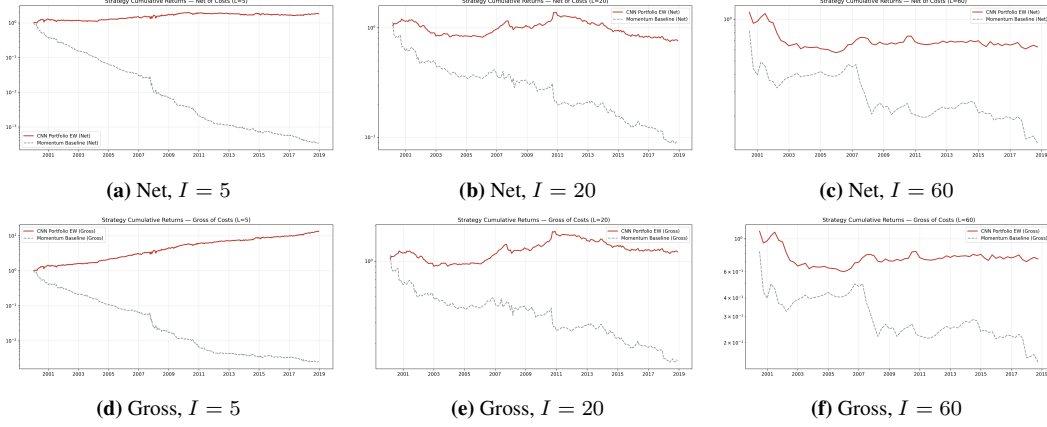
We score every test-set observation with the five-seed ensemble and form ten cross-sectional deciles by predicted “up”-probability at each rebalance date. The high-minus-low (H-L) portfolio is equal-weighted within deciles, rebalanced at horizon-end, and adjusted for proportional transaction costs at 10 bp per side (20 bp round-trip) applied to both legs.

### 5.1 Decile prediction accuracy (Figure 6 reproduction)

At  $I = 5$  the CNN decile annualised returns rise nearly monotonically from 6.62% in D1 to 20.67% in D10 (Figure 3a), while the five-day momentum and moving-average curves are essentially flat across the cross-section. At longer horizons the picture changes: the  $I = 20$  CNN spread between D10 and D1 is mildly positive (11.10% vs. 9.78%; Figure 3b), and at  $I = 60$  the CNN ordering is essentially flat (11.95% in D10 against 13.00% in D1; Figure 3c). The traditional baselines, by contrast, display a U-shaped pattern at  $I = 20$  and  $I = 60$ : both the bottom and top deciles deliver above-average returns, reflecting the well-known concentration of momentum and mean-reversion premia in the extreme deciles.

### 5.2 Comparison with traditional baselines (Figure 7)

Figure 4 shows that at  $I = 5$  the CNN ranking strictly dominates both momentum and moving-average cross across the cross-section, but is itself dominated in absolute level by the short-term reversal baseline whose D10 annualised return reaches 37.78% (driven by the microstructure-style bid-ask bounce on which it relies). At  $I = 20$  the momentum and moving-average curves display the same U-shaped pattern noted above, and the CNN’s monotone ranking is qualitatively closer to



**Figure 5:** Cumulative log returns of the H-L decile spread, net of transaction costs (top row) and gross (bottom row). The  $I = 5$  spread is the only one whose net curve still rises persistently; the  $I = 20$  and  $I = 60$  net curves flatten relative to their gross counterparts because round-trip turnover is in the 84–87% range (Table 1).

**Table 1:** Equal-weighted decile portfolio statistics (Table 1 reproduction). Ret is annualised mean return, Std annualised standard deviation, SR the resulting Sharpe ratio. “H-L” columns are for the long-short decile spread net of 20 bp round-trip transaction costs. Numbers are read from `outputs/pipeline_runs/tables/table1_I{5,20,60}.csv`.

|                                                                  | D1   | D2   | D3   | D4   | D5   | D6   | D7   | D8   | D9   | D10  | H-L   |
|------------------------------------------------------------------|------|------|------|------|------|------|------|------|------|------|-------|
| <b><math>I = 5</math> (weekly rebalance, turnover 0.858)</b>     |      |      |      |      |      |      |      |      |      |      |       |
| Ret                                                              | 0.07 | 0.12 | 0.13 | 0.13 | 0.14 | 0.15 | 0.16 | 0.16 | 0.17 | 0.21 | 0.14  |
| Std                                                              | 0.13 | 0.16 | 0.17 | 0.17 | 0.18 | 0.18 | 0.19 | 0.19 | 0.20 | 0.21 | 0.11  |
| SR                                                               | 0.55 | 0.75 | 0.77 | 0.77 | 0.75 | 0.79 | 0.83 | 0.85 | 0.89 | 1.04 | 1.29  |
| <b><math>I = 20</math> (monthly rebalance, turnover 0.841)</b>   |      |      |      |      |      |      |      |      |      |      |       |
| Ret                                                              | 0.10 | 0.12 | 0.13 | 0.12 | 0.11 | 0.12 | 0.11 | 0.11 | 0.11 | 0.11 | 0.01  |
| Std                                                              | 0.17 | 0.18 | 0.18 | 0.18 | 0.18 | 0.18 | 0.17 | 0.17 | 0.17 | 0.17 | 0.08  |
| SR                                                               | 0.59 | 0.63 | 0.72 | 0.65 | 0.63 | 0.65 | 0.65 | 0.66 | 0.68 | 0.68 | 0.16  |
| <b><math>I = 60</math> (quarterly rebalance, turnover 0.867)</b> |      |      |      |      |      |      |      |      |      |      |       |
| Ret                                                              | 0.13 | 0.13 | 0.13 | 0.14 | 0.13 | 0.12 | 0.13 | 0.13 | 0.12 | 0.12 | -0.01 |
| Std                                                              | 0.22 | 0.20 | 0.20 | 0.19 | 0.20 | 0.19 | 0.18 | 0.18 | 0.17 | 0.16 | 0.11  |
| SR                                                               | 0.60 | 0.65 | 0.67 | 0.71 | 0.65 | 0.67 | 0.69 | 0.73 | 0.73 | 0.77 | -0.11 |

the monthly-frequency pattern they emphasise when comparing CNN deciles to traditional signals. At  $I = 60$  the CNN ranking is nearly horizontal both here and in their Figure 6(c), consistent with the well-documented attenuation of price-trend signals at quarterly horizons.

### 5.3 Cumulative returns

Figure 5 plots the cumulative log return of the H-L spread. The  $I = 5$  net curve climbs steadily because the high-frequency ranking is sharp enough to absorb the transaction-cost drag. At  $I = 20$  and  $I = 60$ , the gross curves are mildly positive but the net curves are substantially attenuated: with turnover at 84.08% and 86.68% per period (Table 1), the 20 bp round-trip cost erases most of the gross spread.

### 5.4 Decile portfolio statistics (Table 1 reproduction)

Table 1 confirms the pattern visible in Figures 3–5: only the  $I = 5$  horizon delivers a robust net H-L spread (annualised SR 1.29). At  $I = 20$  the H-L SR is 0.16, and at  $I = 60$  the spread is mildly negative (−0.11). The monotone increase in within-decile Sharpe ratios at  $I = 60$  (from 0.60 at D1 to 0.77 at D10) shows that the CNN *ordering* is still informative, but the gross spread is too small relative to volatility to survive transaction costs.

**Table 2:** Out-of-sample binary classification metrics for the CNN ensemble.  $N_{\text{test}}$  is the size of the outputs/cache/preds\_I\*.npy vector, i.e. the number of rebalance-day observations scored by the five-seed ensemble.

| Horizon  | Accuracy | AUC-ROC | $N_{\text{test}}$ |
|----------|----------|---------|-------------------|
| $I = 5$  | 49.45%   | 0.5103  | 6,378,408         |
| $I = 20$ | 52.68%   | 0.5242  | 1,441,167         |
| $I = 60$ | 53.70%   | 0.5288  | 453,847           |

**Table 3:** Fama-French five-factor plus momentum spanning regression of the net long-short CNN spread  $r_t^{\text{CNN,LS}} - r_t^f$  on {Mkt-RF, SMB, HML, RMW, CMA, Mom} at the matched rebalance frequency. Alphas are annualised;  $t$ -statistics are reported in parentheses. Numbers come from outputs/pipeline\_runs/tables/ff\_regression\_combined.csv.

|                          | $I = 5$         | $I = 20$         | $I = 60$         |
|--------------------------|-----------------|------------------|------------------|
| $\alpha$ (annualised, %) | +0.28<br>(0.13) | −4.30<br>(−2.23) | −6.65<br>(−3.77) |
| $\beta_{\text{Mkt-RF}}$  | 0.29 (14.24)    | 0.10 (2.21)      | 0.08 (1.24)      |
| $\beta_{\text{SMB}}$     | 0.14 (3.81)     | 0.01 (0.11)      | −0.14 (−1.24)    |
| $\beta_{\text{HML}}$     | 0.13 (3.57)     | 0.02 (0.27)      | 0.23 (2.21)      |
| $\beta_{\text{RMW}}$     | 0.02 (0.41)     | 0.32 (3.48)      | 0.69 (5.24)      |
| $\beta_{\text{CMA}}$     | −0.25 (−4.20)   | −0.11 (−0.98)    | −0.29 (−2.01)    |
| $\beta_{\text{Mom}}$     | −0.03 (−1.51)   | −0.00 (−0.05)    | 0.23 (3.86)      |
| $R^2$                    | 0.34            | 0.07             | 0.63             |
| $N$ periods              | 989             | 226              | 74               |

## 5.5 Classification metrics

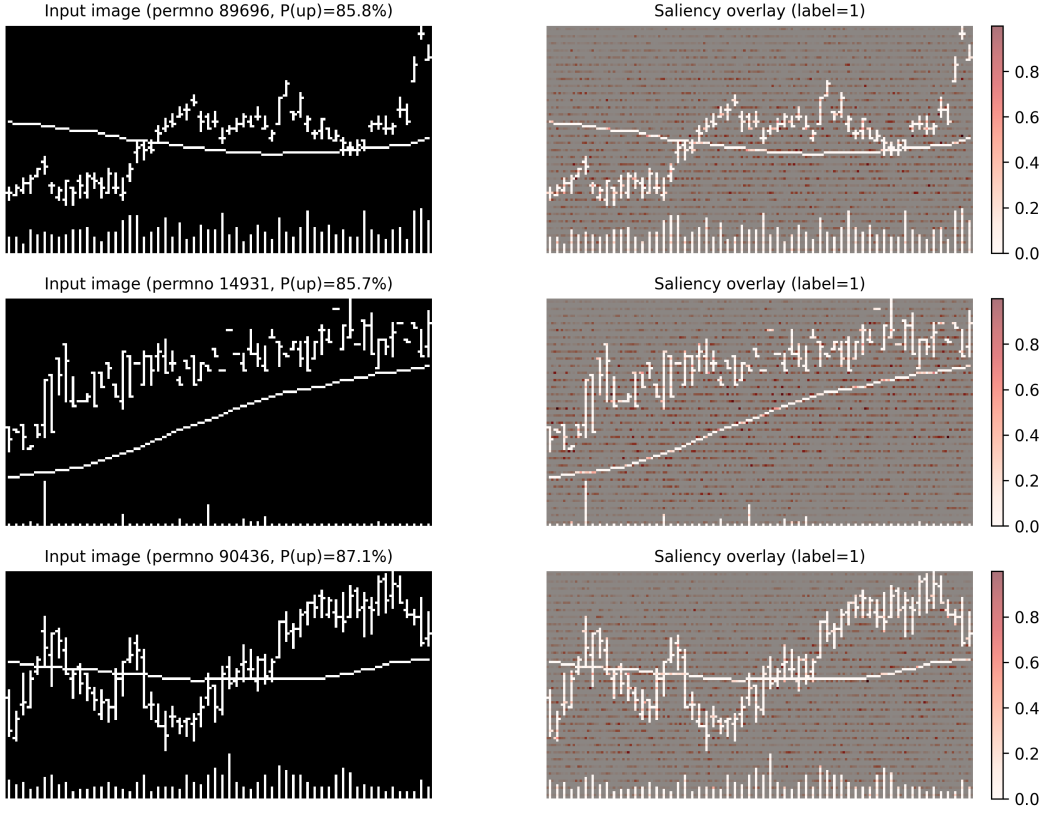
Table 2 reads numbers from outputs/ppt\_images/ppt16\_classification\_metrics.csv. AUC rises with horizon, in line with Jiang et al. [2023], who argue that longer look-back windows expose more learnable structure; accuracy at  $I = 5$  is *below* the 50% benchmark, which is consistent with a left-skewed label distribution (more days lose money in the weekly rebalance universe) rather than a strictly negative signal: the H-L Sharpe of 1.29 shows that the predicted-probability *ranking* is far more informative than the binary forecast.

## 5.6 Fama-French five-factor + momentum spanning

The regression in Table 3 delivers three messages. First, at  $I = 5$  the strategy’s annualised alpha is statistically indistinguishable from zero (+0.28%,  $t = 0.13$ ): the high raw Sharpe is largely captured by factor exposures (significant loadings on Mkt-RF, SMB, HML, and an opposite-signed load on CMA). Second, at  $I = 20$  and  $I = 60$  the alphas are *negative* and significant (−4.30%,  $t = -2.23$ ; −6.65%,  $t = -3.77$ ); the loadings on RMW are positive and large, and at  $I = 60$  the strategy also picks up a positive momentum loading. Third, the very high  $R^2$  at  $I = 60$  (0.63) suggests that the CNN signal at this horizon is essentially a re-weighting of standard style factors rather than an independent source of premia.

## 6 Interpretability: input-gradient saliency

Following Simonyan et al. [2014], we compute the absolute pixel-level gradient of the “up” logit with respect to each input image for three high-confidence positive examples drawn from the  $I = 60$  test window (Figure 6). Across all three examples the saliency mass is concentrated on the right-most 10–20% of the OHLC panel and on volume bars near sharp price moves. This is consistent with a data-driven momentum reading: the CNN essentially learns to up-weight the most-recent observations, which is also the regime in which classical price-trend signals concentrate their power.



**Figure 6:** Input-gradient saliency for three high-confidence positive predictions of the  $I = 60$  ensemble. Left column: the input chart (inverted to a white background for readability). Right column: the same image with the gradient magnitude  $|\partial f_{\text{up}}/\partial x|$  overlaid in red. The network’s attention concentrates on the right-most third of each window (i.e. the most-recent price action) and on volume bars near regime changes.

## 7 Discussion and conclusion

We recover the qualitative shape of Jiang et al. [2023] at  $I = 5$  – a monotone decile ranking, a high gross-of-cost H–L spread that survives transaction costs into a 1.29 net Sharpe ratio, and a decile-by-decile dominance over momentum and moving-average cross. At  $I = 20$  and  $I = 60$  the qualitative ranking is preserved (Table 1, Figure 3), but the H–L spread is too thin to survive costs, and the Fama–French six-factor regression delivers a statistically significant *negative* alpha at both horizons.

Several factors plausibly account for numerical differences relative to Jiang et al. [2023]. (i) Our CRSP-style extract, liquidity filters, and ETF-based calendar construction need not coincide exactly with theirs. (ii) We aggregate five randomly seeded models; their headline results pool a larger ensemble, which can stabilise probability estimates. (iii) Small implementation choices—chart rasterisation, float rounding, and how turnover maps into proportional costs—can move Sharpe ratios and alphas when gross spreads are thin.

The saliency analysis (Figure 6) shows that the  $I = 60$  CNN learns a momentum-like reading rather than a more exotic pattern; this is consistent with a model whose alpha is largely spanned by RMW and Mom. Promising directions for future work include regime-conditioned heads, industry- or size-neutralised label construction, and a mixture-of-experts ensemble that explicitly allocates capacity across horizons.

## References

- Mark M. Carhart. On persistence in mutual fund performance. *Journal of Finance*, 52(1):57–82, 1997.
- Eugene F. Fama and Kenneth R. French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993.
- Eugene F. Fama and Kenneth R. French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015.
- Shihao Gu, Bryan T. Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.
- Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1):65–91, 1993.
- Jingwen Jiang, Bryan T. Kelly, and Dacheng Xiu. (re-)imag(in)ing price trends. *Journal of Finance*, 78(6):3193–3249, 2023.
- Bryan T. Kelly, Seth Pruitt, and Yinan Su. Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501–524, 2019.
- Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Andrew W. Lo, Harry Mamaysky, and Jiang Wang. Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation. *Journal of Finance*, 55(4):1705–1765, 2000.
- Omer Berat Sezer and Ahmet Murat Ozbayoglu. Algorithmic financial trading with deep convolutional neural networks: Time series to image conversion approach. *Applied Soft Computing*, 70:525–538, 2018.
- Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. Deep inside convolutional networks: Visualising image classification models and saliency maps. In *International Conference on Learning Representations (ICLR) Workshop*, 2014.

## A Reproduction artefact map

Every numerical claim in this paper traces back to a file in `final_submit_version/outputs/`:

- Figure 1: produced by `scripts/pipeline_split/export_paper_assets.py`.
- Figure 3, Figure 4: PNGs in `pipeline_runs/figures/`; decile-level CSVs in `pipeline_runs/tables/fig{6,7}_simple_I{5,20,60}_{cnn,ma,mom,reversal}.csv`.
- Figure 5: PNGs in `ppt_images/ppt15_cumulative_returns{,_gross}_L{5,20,60}.png`.
- Table 1: `pipeline_runs/tables/table1_I{5,20,60}.csv`.
- Table 2: `ppt_images/ppt16_classification_metrics.csv` plus shape of `cache/preds_I{5,20,60}.npz`.
- Table 3: produced by `scripts/pipeline_split/export_ff_regression.py`; intermediate file `pipeline_runs/tables/ff_regression_combined.csv`.
- Figure 6: produced by `scripts/pipeline_split/export_saliency_map.py` using the  $I = 60$  checkpoint at seed 42.

## B Hyperparameters and per-block tensor shapes

Table 4 reports the per-block configuration used in the three CNN backbones; all blocks share the `Conv→BatchNorm→LeakyReLU(0.01)→MaxPool(2, 1)` pattern.



**Table 4:** Per-block convolutional configuration. “–” denotes a block that the horizon does not use.

| Block            | $I = 5$              | $I = 20$              | $I = 60$              |
|------------------|----------------------|-----------------------|-----------------------|
| Block 1 channels | $1 \rightarrow 64$   | $1 \rightarrow 64$    | $1 \rightarrow 64$    |
| Block 1 stride   | (1,1)                | (3,1)                 | (3,1)                 |
| Block 1 dilation | (1,1)                | (2,1)                 | (3,1)                 |
| Block 2 channels | $64 \rightarrow 128$ | $64 \rightarrow 128$  | $64 \rightarrow 128$  |
| Block 3 channels | –                    | $128 \rightarrow 256$ | $128 \rightarrow 256$ |
| Block 4 channels | –                    | –                     | $256 \rightarrow 512$ |
| FC input dim     | 15,360               | –                     | 184,320               |
| Approx. params   | $\sim 150\text{k}$   | $\sim 700\text{k}$    | $\sim 3.0\text{M}$    |

## C Computational footprint

The cleaned daily panel cache (`final_submit_version/outputs/cache/cleaned_data_all.pkl`) is  $\sim 11.1$  GB. Per-seed checkpoint sizes are 0.6 MB ( $I = 5$ ), 2.8 MB ( $I = 20$ ), and 11.8 MB ( $I = 60$ ); five seeds per horizon give a total of  $\approx 76$  MB on disk. Training and ensemble inference were performed on a single CUDA GPU (with an MPS fallback for Apple-silicon development); end-to-end execution of the pipeline takes on the order of a few wall-clock hours from the cleaned cache.

## NeurIPS Paper Checklist

### 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract and Section 1 state precisely what we reproduce, with horizon-specific Sharpe ratios, AUCs, and Fama–French alphas; every numerical claim is traceable to a CSV listed in Appendix A.

Guidelines:

- The answer [\[N/A\]](#) means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [\[No\]](#) or [\[N/A\]](#) answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 7 catalogues divergences from the original paper at  $I = 20$  and  $I = 60$ , attributing them to post-publication signal decay, ensemble size, transaction-cost handling, and volume-bar normalisation.

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